Slide Set #9

Asymptotic Approximations Pt. II

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Notes

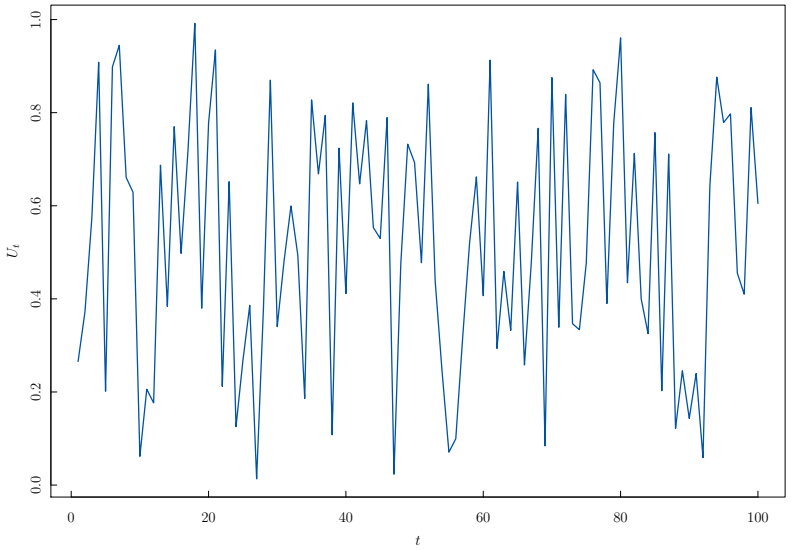
Time series

- Time series: is a stochastic process $\{z_i\}$, with i denoting time.
- Ordering: now the index i is central, if random variables of the sequence are shuffled we obtain a different stochastic process.
- Sample path (trajectory): is a realization of the stochastic process. The fundamental problem in time series is that we can observe a single realization, and usually, only for a finite time period.

Notes

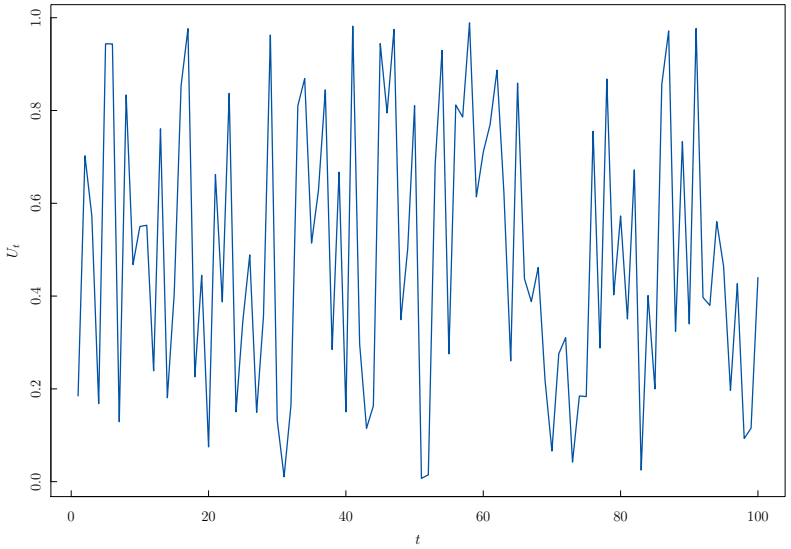
Consider $\{U_t\}$, where $U_t \sim$ Uniform$(0,1)$ for $t = 1, 2, \ldots$

This is a sample path (realization)



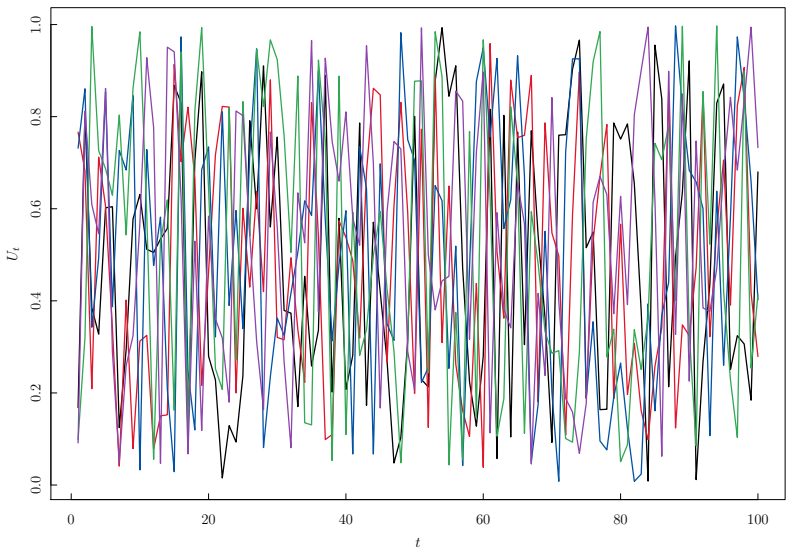
Notes

Another sample path (realization)



Notes

Several sample paths



Notes

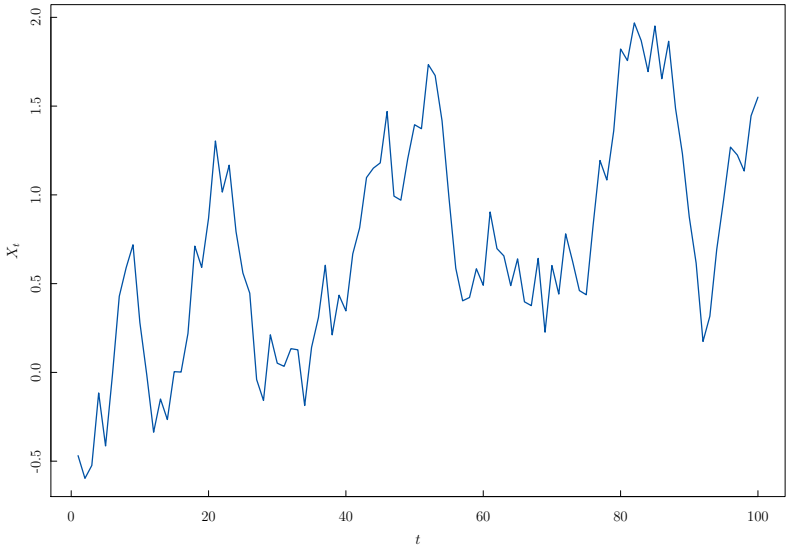
Now consider $\{X_t\}$, for $t = 1, 2, \ldots$ the sequence is generated by

$$X_t = X_{t-1} + \varepsilon_t$$
$$\varepsilon_t \sim \text{Uniform}\left(-\frac{1}{2}, \frac{1}{2}\right)$$
$$X_0 \sim \text{Uniform}\,(-1, 1)$$

Observe: given the initial drawn X_0 this affects all future realizations

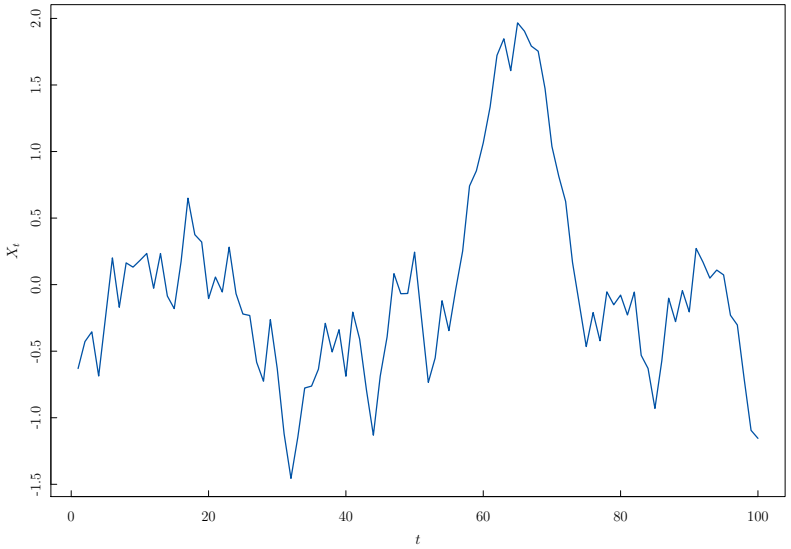
Notes

This is a sample path (realization)



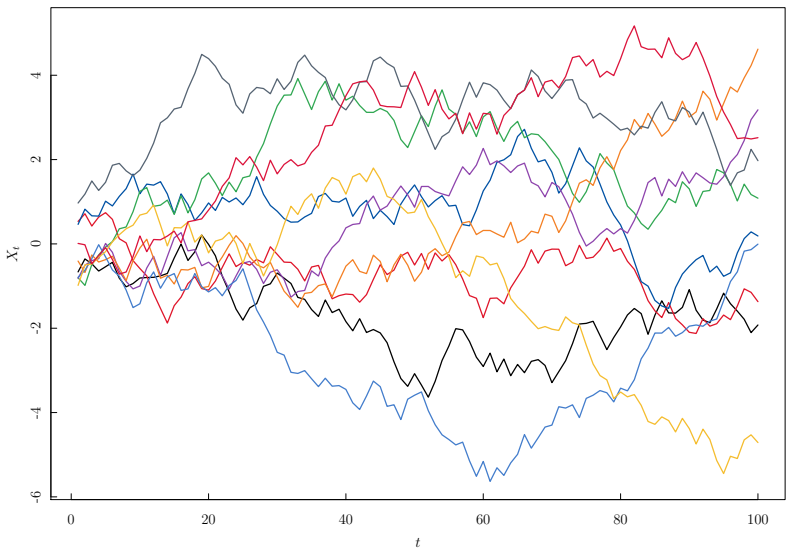
Notes

Another sample path (realization)



Notes

Several sample paths



Notes

In order to do statistics with dependent processes we need two key properties: stationarity and long-run dependence

Stationarity

features of the distribution of z_i (e.g. moments) do not change over time. Essentially it provides a framework in which centrality measures (e.g means) make sense.

Long run independence restrictions

you want that the average of a sufficiently long time series (single sample path) is close enough to that of the underlying distribution. This can be achieved if, at least asymptotically, the process becomes somewhat independent.

Notes

Moments of stochastic processes

Take a scalar process $\{z_i\}$, take a (time) index i, and a lag j

- Mean function: $\mu_i = \mathrm{E}[z_i]$
- Autocovariance function: $\gamma(i, i+j) = \mathrm{Cov}(z_i, z_{i+j})$
- Autocorrelation function: $\rho(i, i+j) = \mathrm{Cor}(z_i, z_{i+j})$
  (also called *serial correlation*)

Issues with mean/variance

when depend on i there are aspect of the distribution of z that are function of i, therefore our ability to predict the future depends on the specific time point i

Relevance of auocovariance/autocorrelation

if there is a sufficient amount of correlation between z_i and z_{i+j} means that we can predict future realization of z. We can do this if the serial-correlation structure doesn't change too much over time.

Notes

Suppose $\boldsymbol{z}_i$ is now of dimension $(K \times 1)$.

Mean function: $\boldsymbol{\mu}_i = \mathrm{E}[\boldsymbol{z}_i]$

Autocovariance function:

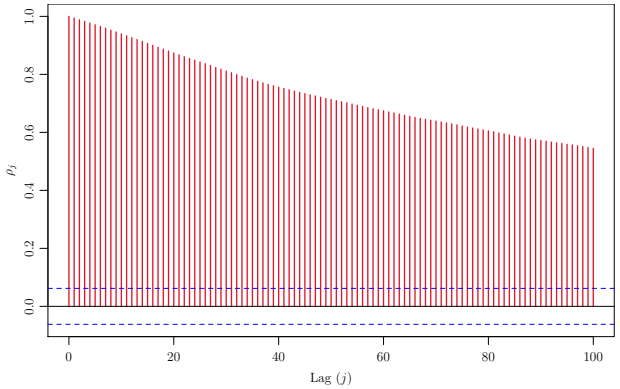
$$\boldsymbol{\Gamma}(i, i+j) = \mathrm{Cov}(\boldsymbol{z}_i, \boldsymbol{z}_{i+j}) \;=\; \mathrm{E}\left[(\boldsymbol{z}_i - \boldsymbol{\mu}_i)(\boldsymbol{z}_{i+j} - \boldsymbol{\mu}_{i+j})'\right]$$

Now it is a $(k \times k)$ matrix. It is essentially the cross-covariance between $\boldsymbol{z}_i$ and $\boldsymbol{z}_{i+j}$.

Notes

The *correlogram* plot is an effective visualizzation device where we plot $\rho(0, j)$ against $j = 0, 1, 2, \ldots,$.

For example here we plot the correllogram of a realization of X_t (previous example)



Notes

Stationarity

Stationarity (strict) means that the probability distribution of $\{\boldsymbol{z}_i\}$ is invariant under shifts in time.

Definition (Stationarity, or strict stationarity)

The stochastic process $\{\boldsymbol{z}_i\}$ is (strictly) stationary if the joint distribution function of $(\boldsymbol{z}_{i_1}, \boldsymbol{z}_{i_2}, ..., \boldsymbol{z}_{i_r})$ is equal to the joint distribution function of $(\boldsymbol{z}_{i_1+j}, \boldsymbol{z}_{i_2+j}, ..., \boldsymbol{z}_{i_r+j})$ for any set of indexes $\{i_1, i_2, ..., i_r\}$ and the lag j

Example: the joint distribution of $(\boldsymbol{z}_1, \boldsymbol{z}_{10})$ is same of that $(\boldsymbol{z}_{101}, \boldsymbol{z}_{110})$, which in turn is equal to that of $(\boldsymbol{z}_{51}, \boldsymbol{z}_{60})$.

- If $\{\boldsymbol{z}_i\}$ is stationary, $\{g(\boldsymbol{z}_i)\}$ is also stationary for any measurable function $g(\cdot)$
- If the vector process $\{\boldsymbol{z}_i\}$ is (strictly) stationary $\implies$ (strict) stationary element-wise. The converse is not true

Notes

Weak stationarity

Definition (weak stationarity, aka covariance stationarity)

A vector process $\{\boldsymbol{z}_i\}$ is covariance stationary if

1. $\mathrm{E}[\boldsymbol{z}_i] = \boldsymbol{\mu}$ for all i;
2. $\mathrm{Var}[\boldsymbol{z}_i] < +\infty$ for all i
3. $\mathrm{Cov}[\boldsymbol{z}_i, \boldsymbol{z}_{i+j}] = \boldsymbol{\Gamma}_j$ for all i and $j \in \mathbb{Z}$

Main features

- the mean and variance do not depend on i
- the linear dependence between $\boldsymbol{z}_i$ and $\boldsymbol{z}_{i+j}$ only depends on j whatever the time point considered i. Example: the correlation between GDP_{1980} and GDP_{1979}, is the same as between GDP_{2021} and GDP_{2020}

Implications

- strict stationarity + finite variance $\implies$ weak stationarity
- weak stationarity + $\boldsymbol{z}_i \sim$Normal $\implies$ strict stationarity.

Notes

Symmetry

since autocovariance only depends on the lag j, than $\gamma(j) = \gamma(-j)$ because

$$\gamma(j) = \mathrm{Cov}[z_i, z_{i+j}] = \mathrm{Cov}[z_{i-j}, z_i] = \gamma(-j)$$

For vector processes one has $\boldsymbol{\Gamma}_j = \boldsymbol{\Gamma}'_{-j}$

Unconditional variance

is given by $\mathrm{Var}[z_i] = \mathrm{Cov}[z_i, z_{i+0}] = \gamma(0)$. For vector processes $\boldsymbol{\Gamma}_0 = \mathrm{Var}[\boldsymbol{z}_i]$.

Autocovariance matrix of weak stationary scalar process

is the variance matrix of the random vector $(z_i, z_{i+1}, z_{i+2}, \ldots, z_n)'$. The element of its l-th row and q-th column is $\mathrm{Cov}[z_{i+l}, z_{i+q}] = \gamma(|l-q|)$

$$\begin{pmatrix} \gamma_0 & \gamma_1 & \gamma_2 & \cdots & \gamma_{n-1} \\ \gamma_1 & \gamma_0 & \gamma_1 & \cdots & \gamma_{n-2} \\ \gamma_2 & \gamma_1 & \gamma_0 & \cdots & \gamma_{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \gamma_{n-1} & \gamma_{n-2} & \gamma_{n-3} & \cdots & \gamma_0 \end{pmatrix}$$

Notes

since for scalar processes $\rho(j) = \gamma(j)/\gamma(0)$, the corresponding autocorrelation matrix is

$$\begin{pmatrix} 1 & \rho_1 & \rho_2 & \cdots & \rho_{n-1} \\ \rho_1 & 1 & \rho_1 & \cdots & \rho_{n-2} \\ \rho_2 & \rho_1 & 1 & \cdots & \rho_{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho_{n-1} & \rho_{n-2} & \rho_{n-3} & \cdots & 1 \end{pmatrix}$$

Observe

- autocovariance(autocorrelation) matrix of weak stationary process gives the important overview of all the linear dependence between shifted(lagged) terms of z
- the picture it provides is exhaustive because for weak stationary process the linear dependence only depends on the shift(lag), and it does not depend on the time point considered

Notes

A crucial class of weakly stationary processes is the following

Definition (White noise)

The process $\{\boldsymbol{z}_i\}$ is white noise if for all i

- $\mathrm{E}[\boldsymbol{z}_i] = \mathbf{0}$
- $\mathrm{Var}[\boldsymbol{z}_i] = \boldsymbol{\Gamma}_0$
- $\mathrm{Cov}\,[\boldsymbol{z}_i, \boldsymbol{z}_{i+j}] = \mathbf{0}$ for all $j \neq 0$

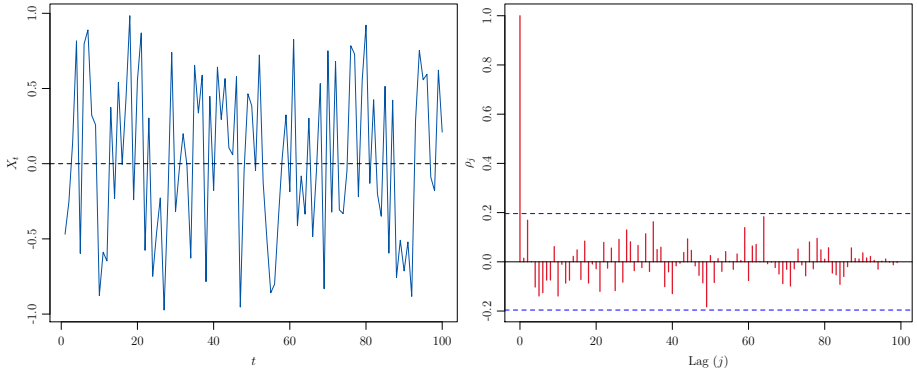
Observe: we do not require independence, just linear independence. That's why iid sequences are called independent white noise

A white noise processes is **not necessarily** strictly stationary

Notes

White Noise (Weak and Strict Stationary)

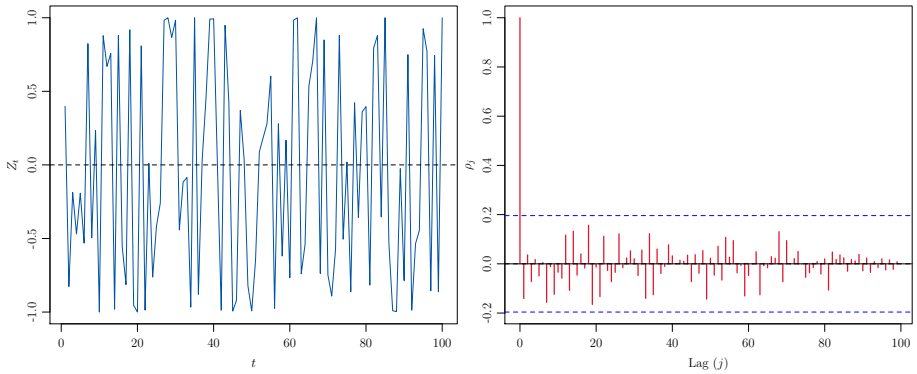
$\{X_t\}$ is an iid sequence with $X_t \sim$Uniform$(-1, 1)$



Notes

White Noise (Weak but not Strict Stationary)

$\{Z_t = \cos(tu_t)\}$, where $\{u_t\}$ is an iid sequence with $u_t \sim$Uniform$(0, 2\pi)$



Notes

Ergodicity

Let S be the outcome space, $z_i(s)$ is the value of the process at time i when $s \in S$ happens. Suppose we want to estimate moments of $h(z_i)$, that is averages of $h(z_i(s))$ for some function $h(\cdot)$

In order to use the unique sample path we would like statements like

$$\frac{1}{n}\sum_{i=1}^{n} h(z_i(s)) \longrightarrow \mathrm{E}[h(z(s))] \quad \text{(p,as, or ms sense)}$$

Why? We would like to use the "long run time average" for estimating the underlying mean (moment across S).

However, in order to have this LLn-type results we need to have some *dependence restrictions*. Why?

Notes

If in the infinite future z still depends on what happened in the early times

- the sample path we are using is bonded too strongly to the beginning, and therefore it may not well represent all other possible sample paths
- if the dependence doesn't vanish after having observed z for so long, the variance of the sum of the lhs would explode and any LLN-type statement would be hopeless

Ergodicity is an asymptotic independence restriction. Definition of ergodic sequences is a technical monster. Let's put it simple.

Notes

Definition (Ergoticity, heuristic definition)

$\{\boldsymbol{z}_i\}$ is an ergodic process if $\boldsymbol{z}_i$ and $\boldsymbol{z}_j$ are independent whenever $|i-j|$ is large enough.

If $\{\boldsymbol{z}_i\}$ is ergodic stationary $\{g(\boldsymbol{z}_i)\}$ is also ergodic stationary for any measurable $g(\cdot)$. This is crucial to prove consistency results.

Theorem (Birkhoff-Khinchin, 1931)

Let $\{\boldsymbol{z}_i\}$ be a stationary ergodic process with $\mathrm{E}[\boldsymbol{z}_i] = \boldsymbol{\mu} < \infty$, then

$$\bar{\boldsymbol{z}}_n = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{z}_i \xrightarrow{\text{as}} \boldsymbol{\mu}$$

In most cases one can prove ergodic-type theorems under simpler conditions (eg. jth order autocovariances vanish at suitable speed).

Notes

Information and Martingales

How do we collect information? Your data set on your computer accumulates samples as you collect more and more data. So our information set is something that *evolves*.

Let $\mathcal{I}_{2017}$ be the *information set* in 2017: it contains the GDP records for 2017 and all the previous years.

Definition (Filtrations)

The family $\{\mathcal{I}_i\}$ is a filtration if $\mathcal{I}_t \subset \mathcal{I}_s$ whenever $t \leq s$.

Definition (Processes adaptive to filtrations)

A process $\{\boldsymbol{z}_i\}$ adapts to the filtration $\{\mathcal{I}_i\}$ if the information generated by $\boldsymbol{z}_i$ is contained in $\mathcal{I}_i$ for every i.

Notes

From now onward:

- $\mathcal{I}_i$ = information set available at time i
- the family $\{\mathcal{I}_i\}$ is a filtration, this formalizes the concept that information grows

Definition (Martingale)

A stochastic process $\{\boldsymbol{z}_i\}$ is a martingale if

1. it adapts to a filtration $\{\mathcal{I}_i\}$
2. $\mathrm{E}[|\boldsymbol{z}_i|] < \infty$ for all i
3. $\mathrm{E}[\boldsymbol{z}_i \,|\, \mathcal{I}_{i-1}] = \boldsymbol{z}_{i-1}$ for all i

- Property (3) $\implies$ based on your GDP records up to 2016 your expectation for GDP_{2017} is the GDP_{2016}
- *Hall's martingale hypothesis* (consuption smoothing): aggregate consumption is a martingale wrt information on GDP, money supply, etc.

Notes

An important martingale process is the *random walk process*

Definition (Random walk)

Let $\{\boldsymbol{x}_i\}$ be an iid sequence (independent white noise). The random walk process $\{\boldsymbol{z}_i\}$ is given by the cumulative sum $\boldsymbol{z}_i = \sum_{k=1}^{i} \boldsymbol{x}_k$.

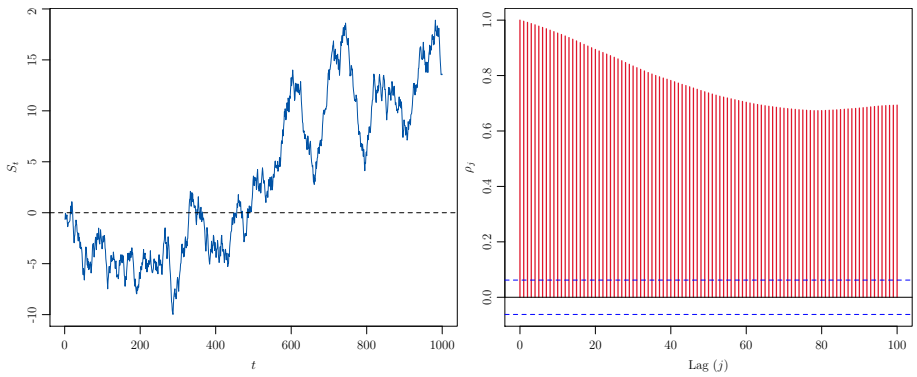
First differences on a random walk give independent white noise:
$\boldsymbol{z}_i - \boldsymbol{z}_{i-1} = \boldsymbol{x}_i$

A random walk is not stationary

To see consider an example: define $S_t = \sum_{i=1}^{t} u_t$, where $\{u_t\}$ is an iid sequence with $u_t \sim$Uniform$(-1, 1)$

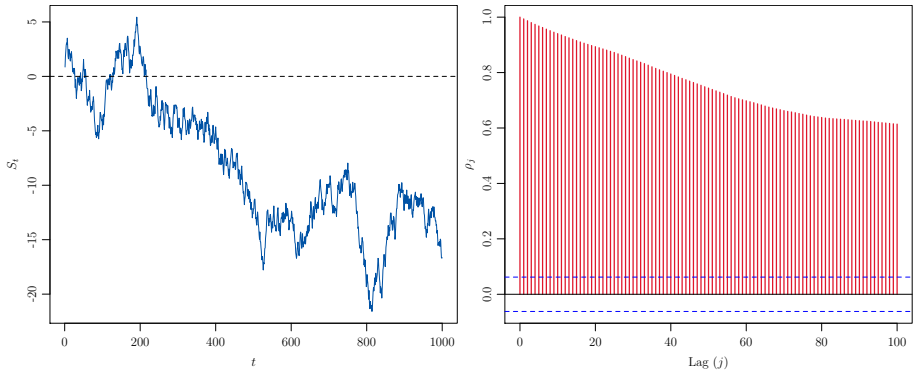
Notes

Sample Path 1



Notes

Sample Path 2



Notes

Definition (Martingale difference sequence, MDS)

A vector process $\{\boldsymbol{g}_i\}$ is a MDS process if

1. it adapts to a filtration $\mathcal{I}_i$
2. $\mathrm{E}[\boldsymbol{g}_i] = \mathbf{0}$ for all i
3. $\mathrm{E}[\boldsymbol{g}_i|\mathcal{I}_{i-1}] = \mathbf{0}$ for all $i \geq 2$.

It can be easily shown that

- first differences of a martingale produces MDS
- independent white noise is a MDS (why?)

The next two properties play a crucial role for many asymptotic results.

Notes

Proposition

The cumulative sum of a MDS $\{\boldsymbol{g}_i\}$ is a martingale $\{\boldsymbol{z}_i\}$.

Proof. By definition $\boldsymbol{z}_i = \boldsymbol{g}_1 + \boldsymbol{g}_2 + \ldots, +\boldsymbol{g}_i$, hence $\boldsymbol{z}_i - \boldsymbol{z}_{i-1} = \boldsymbol{g}_i$.

Observe that $\{\boldsymbol{g}_i\}$ adapts to the filtration $\{\mathcal{I}_i\}$. Hence, the information generated by $\boldsymbol{g}_{i-j}$ is contained in $\mathcal{I}_{i-1}$ for all $j \geq 1$, and therefore

$$\mathrm{E}[\boldsymbol{g}_{i-j}|\mathcal{I}_{i-1}] = \boldsymbol{g}_{i-j}$$

We can now compute

$$\begin{aligned}
\mathrm{E}[\boldsymbol{z}_i \,|\, \mathcal{I}_{i-1}] &= \mathrm{E}[\boldsymbol{g}_i + \boldsymbol{g}_{i-1} + \ldots + \boldsymbol{g}_1 \,|\, \mathcal{I}_{i-1}] \\
&= \mathrm{E}[\boldsymbol{g}_i \,|\, \mathcal{I}_{i-1}] + \mathrm{E}[\boldsymbol{g}_{i-1} \,|\, \mathcal{I}_{i-1}] + \ldots + \mathrm{E}[\boldsymbol{g}_1 \,|\, \mathcal{I}_{i-1}] \\
&= \mathrm{E}[\boldsymbol{g}_i \,|\, \mathcal{I}_{i-1}] + (\boldsymbol{g}_{i-1} + \ldots + \boldsymbol{g}_1) \\
&= \boldsymbol{z}_{i-1}
\end{aligned}$$

This completes the proof ■

Notes

Proposition

Let $\{\boldsymbol{g}_i\}$ be a MDS, then $\mathrm{E}[\boldsymbol{g}_i \,|\, \boldsymbol{g}_{i-j}] = \mathbf{0}$

Proof. Note that $\boldsymbol{g}_{i-j} \subset \mathcal{I}_i$. "Smallest information set wins" $\implies$ $\mathrm{E}[\mathrm{E}[\boldsymbol{g}_i \,|\, \mathcal{I}_{i-1}] \,|\, \boldsymbol{g}_{i-j}] = \mathrm{E}[\boldsymbol{g}_i \,|\, \boldsymbol{g}_{i-j}] = \mathbf{0}$ ■

Proposition

Let $\{\boldsymbol{g}_i\}$ be a MDS, then $\mathrm{Cov}[\boldsymbol{g}_i, \boldsymbol{g}_{i-j}] = \mathbf{0}$ for all i and $j \neq 0$.

Proof: Since $\mathrm{E}[\boldsymbol{g}_i] = \mathbf{0}$ it suffices to show that $\mathrm{E}[\boldsymbol{g}_i \boldsymbol{g}_{i-j}'] = \mathbf{0}$.

$$\begin{aligned}
\mathrm{E}[\boldsymbol{g}_i \boldsymbol{g}_{i-j}'] &= \mathrm{E}[\mathrm{E}[\boldsymbol{g}_i \boldsymbol{g}_{i-j}' \,|\, \boldsymbol{g}_{i-j}]] && \text{(iterated expectations)} \\
&= \mathrm{E}[\mathrm{E}[\boldsymbol{g}_i \,|\, \boldsymbol{g}_{i-j}] \boldsymbol{g}_{i-j}'] && \text{(pull out what's known)} \\
&= \mathbf{0} && \text{(previous property)}
\end{aligned}$$

■

Notes

Strength of independence

We will focus on zero mean, covariance stationary processes. These are sorted from most to less *restricted:*

zero mean, finite variance iid sequences (aka independent white noise)
absence of any form of dependence, zero unconditional mean, zero conditional mean.

stationary MDS with finite variance
allow nonlinear dependence, zero unconditional mean, zero conditional mean, need to adapt to filtrations. A famous example is the ARCH process that revolutionized the financial literature.

white noise
allow nonlinear dependence, zero unconditional mean

Notes

CLT for MDS

Theorem (CLT for ergodic MDS)

Let $\{\boldsymbol{g}_i\}$ be a stationary ergodic MDS with $\mathrm{E}[\boldsymbol{g}_i\boldsymbol{g}_i'] = \boldsymbol{\Sigma}$, let $\overline{\boldsymbol{g}} := \frac{1}{n}\sum_{i=1}^n \boldsymbol{g}_i$

$$\sqrt{n}\overline{\boldsymbol{g}} = \frac{1}{\sqrt{n}}\sum_{i=1}^n \boldsymbol{g}_i \xrightarrow{\mathsf{d}} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$$

- $\boldsymbol{\Sigma}$ is the unconditional variance: since $\mathrm{E}[\boldsymbol{g}_i] = \mathbf{0}$, $\mathrm{Var}[\boldsymbol{g}_i] = \boldsymbol{\Sigma}$
- the theorem is crucial to approximate the distribution of partial sums of serially uncorrelated processes
- ergodicity assumption is a strong long-run independence assumption. It can be weakened to obtain alternative CLTs.

Notes

Sequence of estimators

Let $\hat{\boldsymbol{\theta}}_n$ be the estimator of $\boldsymbol{\theta}$ at sample size n, and assume that you can take a *sequence of sampling experiments* with increasing n.

Asymptotic estimation error
we look at the sequence $\left\{\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}\right\}$

Asymptotic bias
we look at $\lim_{n\to\infty} \mathrm{E}[\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}]$

Consistency
$\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta} \xrightarrow{\mathsf{p}} \mathbf{0}$, that is $\hat{\boldsymbol{\theta}}_n \xrightarrow{\mathsf{p}} \boldsymbol{\theta}$
(consistency can be based also on $\xrightarrow{\mathsf{as}}$ or $\xrightarrow{\mathsf{ms}}$ notion).

$\sqrt{n}$-consistency
$\sqrt{n}\left\{\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}\right\} \xrightarrow{\mathsf{p}} \mathbf{0}$

Asymptotic normality
$\sqrt{n}\left\{\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}\right\} \xrightarrow{\mathsf{d}} \mathcal{N}(\mathbf{0}, \boldsymbol{V})$

Notes

Remark on consistent and asymptotically normal estimators

Consider a sequence of estimators $\{\hat{\boldsymbol{\theta}}_n\}$ fulfilling the CLT

$$\sqrt{n}\left\{\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}\right\} \xrightarrow{\text{d}} \mathcal{N}(\mathbf{0}, \boldsymbol{V})$$

⚠ Based on the CLT above, it t is often concluded that

$$\hat{\boldsymbol{\theta}}_n \sim \mathcal{N}\left(\boldsymbol{\theta}, \frac{\boldsymbol{V}}{n}\right)$$

But the previous statement is wrong! In fact, $\boldsymbol{V}/n \to 0$. Therefore, in the limit $\boldsymbol{\theta}_n$ has a degenerate distribution... Otherwise, you won't get consistency!

CLT convergence rate

It turns out that the limit is reached at the magical speed of $c_n = \sqrt{n}$, this is the converge rate of the CLT

n sufficiently large

We take a large n, but slightly before it is so large that $\boldsymbol{V}/n \to 0$.

Notes

Asymptotic approximate distribution

For n sufficiently large we can make the correct statement that the distribution of $\hat{\boldsymbol{\theta}}_n$ is quite close to $\mathcal{N}\left(\boldsymbol{\theta}, \frac{\boldsymbol{V}}{n}\right)$, and we write

$$\hat{\boldsymbol{\theta}}_n \overset{\text{a}}{\approx} \mathcal{N}\left(\boldsymbol{\theta}, \frac{\boldsymbol{V}}{n}\right)$$

Asymptotic variance of $\hat{\boldsymbol{\theta}}_n = \mathrm{AVar}(\hat{\boldsymbol{\theta}}_n)$

it is meant to be the var matrix of the scaled-centered sequence, that is

$$\mathrm{AVar}(\hat{\boldsymbol{\theta}}_n) = \boldsymbol{V}$$

$\mathrm{AVar}(\hat{\boldsymbol{\theta}}_n)$ is the limiting var matrix of $\sqrt{n}\left\{\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}\right\}$

Approximate variance of $\hat{\boldsymbol{\theta}}_n$

it is meant to be the variance of $\hat{\boldsymbol{\theta}}_n$ *approximated* at n *sufficiently large*

$$\mathrm{Var}[\hat{\boldsymbol{\theta}}_n] \approx \frac{\boldsymbol{V}}{n} = \frac{\mathrm{AVar}(\hat{\boldsymbol{\theta}}_n)}{n}$$

Notes